BST Vol 8 Ch 5 — Symmetries + Noether's Theorem: T2473-T2475 Conservation Laws (v0.3, Wave 3)

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2026-05-23 Saturday

Vol 8 Chapter 5 — Symmetries + Noether's Theorem

Headline result

Noether's theorem (1918) on substrate D_IV^5 symmetry yields conservation laws (T2473-T2475, Lyra Friday SP-31 #279 closure):

- T2473 Energy Conservation via SO_0(5,2) time-translation invariance + Casimir centrality
- T2474 Momentum Conservation via coset translation-direction invariance
- T2475 Charge Conservation via SO(2) factor invariance

This is THE BEST chapter in Vol 8 — substrate-derivation rigorous via Lyra's Friday theorem batch.

Substrate mechanism

Per Lyra T2473-T2475 + Elie Toy 3504 verification PASS 6/6: - H_sub = Casimir on $L^2(D\_IV^5; L\_\lambda)$; central in U(so(5,2)) - T_E, T_P translation generators in m $\subset$ so(5,2)/(so(5)$\oplus$so(2)) - [H_sub, T_E] = [H_sub, T_P] = 0 → energy + momentum conserved - Q = -i·dπ(J_{SO(2)}) (T2470); SO(2) $\subset$ K isotropy → [H_sub, Q] = 0 → charge conserved - Noether 1918 standard differential geometry argument

Cross-volume dependencies

- T2473-T2475 (Lyra Friday SP-31 #279)
- T2470 (Lyra Friday charge Q)
- Vol 0 Ch 8 (Conservation Laws)
- Toy 3504 verification PASS 6/6

K-audit anchor

K233 (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1

> Energy is conserved — it can change form but never disappear. Same for momentum and electric charge. BST predicts these conservation laws come from substrate symmetries: time-translation gives energy, space-translation gives momentum, gauge-invariance gives charge.

Level 2

Noether's theorem: every continuous symmetry → conserved current. Time-translation symmetry → energy; space-translation → momentum; gauge symmetry → charge.

Per Lyra Friday T2473-T2475: substrate SO_0(5,2) symmetry yields all three classical conservation laws via Casimir centrality. Substrate makes Noether's theorem substrate-natural.

Level 3

T2473 (energy conservation): H_sub = Casimir on $L^2(D\_IV^5;\ L\_\lambda)$; time-translation generator T_E ∈ m ⊂ so(5,2)/(so(5)⊕so(2)). [H_sub, T_E] = 0 via Casimir centrality in U(so(5,2)). Noether → E conserved.

T2474 (momentum conservation): coset translation generators T_P ∈ m. Same Casimir centrality argument → momentum components conserved.

T2475 (charge conservation): Q = -i·dπ(J_{SO(2)}) generator (T2470). SO(2) ⊂ K isotropy → [H_sub, Q] = 0 → charge conserved.

Per Cal #21 dual-gate: EMPIRICAL PASS (conservation laws observed) + MECHANISM PASS (Noether + Casimir centrality + SO(2) isotropy). D-tier RATIFIED.

Per Toy 3504 cross-CI verification PASS 6/6 (Elie Friday EOD).

Bibliography

1. E. Noether (1918): Noether's theorem.
2. Lyra T2473-T2475 (Friday SP-31 #279).
3. Toy 3504 (Elie Friday): cross-CI verification.

— Elie, Vol 8 Ch 5 v0.3, 2026-05-23 Saturday